

Forest cover dynamics (1998 to 2019) and prediction of deforestation probability using binary logistic regression (BLR) model of Silabati watershed, India

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ABSTRACT

Forest is a renewable resource and an imperative part of the total environment. But in the last few decades, forest areas are being highly converted and depleted due to excessive anthropogenic stress particularly for modern societal development. The magnitude of deforestation has been exaggerated particularly in tropical and sub-tropical developing countries of the world. The main objective of the study is to identify the probable areas of deforestation along with the analysis of forest cover dynamics within Silabati watershed of India applying Binary Logistic Regression Model. The forest cover map (1998 and 2019) has been extracted with the help of satellite imageries which have been obtained from USGS. The remote sensing techniques coupled with multivariate statistical analysis (Binary Logistic Regression Model) have been used to prepare probable deforestation areas within the watershed. The statistical analysis also signifies the correlation among dependent and independent variables. The model shows that the middle and lower middle part of the watershed is highly vulnerable (approximate 24%) whereas the upper basin area is less vulnerable (approximate 48%). The current study reflects that high population density, road construction and initiation of cash crops are responsible principal factors behind probable deforestation. The model is validated by the receiver operating characteristic (ROC) curve and the Area Under Curve (AUC) method and it reflects around 80% overall accuracy of this model. This model is very rational and effective for policy makers and conservators before the execution of any sustainable forest conservation plan.

1. Introduction

Forest means the large expansion or large territory that is mainly dominated by various types of trees and shrubs. Food and Agricultural Organisation (FAO) has systematically defined that the land has at least 0.5 hectare areas; minimum tree height is 5 m along with 10% canopy cover. The definition of forest varies from country to country through the evolution of time. The Forest Survey of India (FSI) has also set a different guideline about the definition of forest. According to FSI, the forest is defined as minimum 1 hectare of land area which is covered by more than 10% tree canopy cover. According to Global Forest Resource Assessment (GFRA), the tropical region is being faced highest rate of deforestation related problems mainly due to very high population density along with high human interference over nature and natural resources in compare with other climatic regions in the world. Many studies illustrated that the forests are being degraded due to anthropogenic stresses

over the decades. The huge anthropogenic stress as well as rapid transformation of forest cover is a serious concern for the health of the total environment along with researchers as it provides us many indispensable resources and also takes a great role in global carbon balance and global carbon emission (van der Werf et al., 2009). Forest land may create an equilibrium condition in the environment as well as broadly in the eco-system, but now some illegal and unscientific activities destruct the forest lands in a large scale day after day (Kayet et al., 2016). Due to human stress and many developmental activities such as road, railway, hydropower, dam construction etc. also lead to convert a dense forest region into a sparse or isolated forest region. In the recent years, continental southeast Asia has started different activities like timbering and varieties of agriculture and as a result the annual deforestation rate (during 2000 to 2015 2.6 million ha land deforested) was quite high (Stibig et al., 2014). South East Asian countries mainly the people of tropical developing countries have highly engaged in forest clearing and destructing activities for the execution of different developmental activ-

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ities (Kaimowitz and Angelsen, 1998; Laurance et al., 2009). Not only the developmental activities but also the forest regions are depleting due to forest fire, timber marketing, and conversion of forest land into agricultural lands that make an environmental instability (Gayen and Saha 2018). The continuous human encroachment along with developmental activities over forest lands aggravate the high magnitude of forest land depletion (Becek and Odihi, 2008), and these activities have also been taken the responsibility of natural habitats fragmentation (Sun and Southworth, 2013). The term forest cover dynamics exhibit the rate, pattern, spatial distribution, change of forest land area due to different human induced factors along with global climatic dynamics (Kumar et al., 2014). The per capita forest land in India is 0.10 ha meanwhile the world average is 1.0 ha that replicates the poorest condition of forest in India in terms of per capita forest lands (Liki 2020). Similarly, India has almost 5400 endemic species and 17,000 flowering plants that make India a mega-biodiversity nation (Hajra and Mudgal 1997). Champion and Seth have classified Indian forest region into 16 major and 221 minor forest categories based on different physiographic and climatic condition (Champion and Seth 1968). Forest Survey of India (FSI) has further reclassified the Indian forest into 16 major and 178 minor forest groups. It has been proved that the rapid rate of urbanization is a key factor of deforestation. In case of Ranchi (India), the area has been reduced by 0.95 km²/year from 1996 to 2008 (Bahugunna et al., 2016). With the advancement of science and technology, the remote sensing technique can detect spatial health of forest condition and rate of deforestation (Guhathakurta and Roy 2000). According to 1986 forest policy a Remote Sensing (RS) and Geographic Information System (GIS) cell should be installed in all states over the India to monitor the forest degradation rate. The progression of science and technology along with emergence of RS-GIS techniques (different multi spectral and hyper spectral images) have been widely used to audit the forest health and rate of deforestation in the tropical forest areas (Azizi, 2008). Remote Sensing techniques is an important discipline in the recent days which also detect and predict the spatial objects within the earth surface but it is more feasible when it would be coupled with other highly acceptable statistical techniques (Saha et al., 2020). Binary Logistic Regression model has been applied in different fields like landslide susceptibility mapping, hazard zonation mapping, deforestation probability mapping, ground water potential mapping etc. (Ozdemir 2011). In the recent years, enormous works have been done on demarcation of deforestation probability zones using the RS and GIS techniques along with the advancement of scientific techniques. Deforestation probability zonation may also predict or quantify the susceptibility zones of forest lost and this may also help the policy makers to take proper management about this current problem (Srivastava et al., 2002; Nandy et al., 2007). Statistical analysis, spatial modelling, machine learning modelling and common parametric model (BLR model) have been accepted for deforestation susceptibility zonation mapping (Dlamini, 2016; Aguilar-Amuchastegui et al., 2014). In this study, authors tried to determine the deforestation probable zones in more accurately and the scientific study has been also taken ample number of validation samples which make the scientific study more relevant in present day context. The study has been successfully completed considering some limitations like assumption of linearity between the dependent variable and independent variables and preference of discrete functions. The main objectives of the present study are i) to identify the deforested areas within the Silabati watershed ii) to define the deforestation probability zones by applying the multivariate statistical approach (Binary Logistic Regression model) within Silabati watershed.

2. Study area

River Silabati originates from Hura block, Purulia district, West Bengal and it takes south westerly direction. Near Ghatal of Paschim Mednapore it enters into river Dwarakeswar. Rupnarayan is the combined flow of Dwarakeswar and Silabati, and finally it falls into river Hooghly. The study area extends between 22°36' to 23°12' North latitude and

86°38' to 87°43' East longitude with an area of approximately 3390 sq.km (Fig. 1). The flow of river is restricted within the three districts of West Bengal namely Purulia, Bankura and Paschim Mednapore and according to Koppen's classification the basin area is under the sub-humid tropical climatic zone with an annual rainfall of 100 to 150 cm and almost 30 °C average temperature. The upper and the lower basin area is characterized by very undulating surface and plain topography with isolated lateritic soil pockets respectively (Shit and Maity 2012). Silabati watershed has different hydro-geomorphic characteristics where the relative height greatly decreases from source region to confluence zone with a highest altitude 1368 m and lowest altitude 8 m. The lower part of the basin is more suitable for human construction in compare with the upper part and some important municipality areas like Chandrakona, Ghatal, Keshpur, Garhbeta are located within the lower part of the basin. The upper part of the basin is dominated by gneiss and schist (unclassified crystalline rocks) while the lower part is surrounded by younger alluvium rocks due to the gradual accumulation of quaternary alluvium (Mahala 2019). The lower portion of the basin area is an extensive flood plain area with some populous places of Paschim Mednapore district such as Ghatal, Khirpai, Mulgram, Chandrakona, Surapur etc. Ghatal is the very high flood prone pocket and extensive agriculture is being practiced on this flood plain region along with high population density. These economic activities with high rate of settlement expansion lead the high probability of deforestation in the middle and lower part of watershed (Mahala, 2019). Almost the entire Paschim Mednapore district is situated within the lower part of the watershed and the district shows high population density along with high growth rate. In recent years, people prefer to cultivate different types of cash crops such as oil seed; pulses etc., along with animal husbandry which accelerate the magnitude of deforestation at the proximity of nodal growth points.

3. Materials and method

3.1. Database

Deforestation is mainly a man-made practice but sometimes it is related with natural process. For the preparation of deforestation probability mapping, eight explanatory factors have been considered (distance from river, agricultural density, altitude, settlement density, forest density, population density, distance from road and slope). Different types of multi spectral satellite data such as Landsat 8 (OLI) and Landsat 5 (TM), Digital Elevation Model (30mts), Google Earth data, population data (Census of India, 2011) have been used here for the depiction of deforestation probability mapping (Tables 1 and 2). The satellite images and Digital Elevation Model have been culled from United State Geological Survey website (<https://earthexplorer.usgs.gov/>). Satellite images are further converted into Universal Transverse Mercator (UTM) projection from geographical coordinate system for calculating the area and better extraction of data. WGS84 datum system has been considered during the time of image processing. Atmospheric correction has been done in Eardas Imagine software (Version: 16.5) for better representation of satellite images and data acquisition. Meanwhile, the proper band combinations have been chosen and images are properly enhanced by Eardas Imagine software for preparation of different thematic maps. Slope, altitude, distance from river data have been derived from the digital elevation model which retrieved from USGS whereas data of population and the road layer have been obtained from census of India (2011) and Google Earth images respectively. Similarly, the agriculture, settlement, forest density data are extracted from the post classification of 2019 OLI (Operational Land Imager) image. The detailed description of the data has been enlisted below.

3.2. Determination of dependent variables: deforested zones demarcation

In this study, deforested zones are considered as a dependent variable. At first the forest zones or forest maps of 1998 and 2019 have been

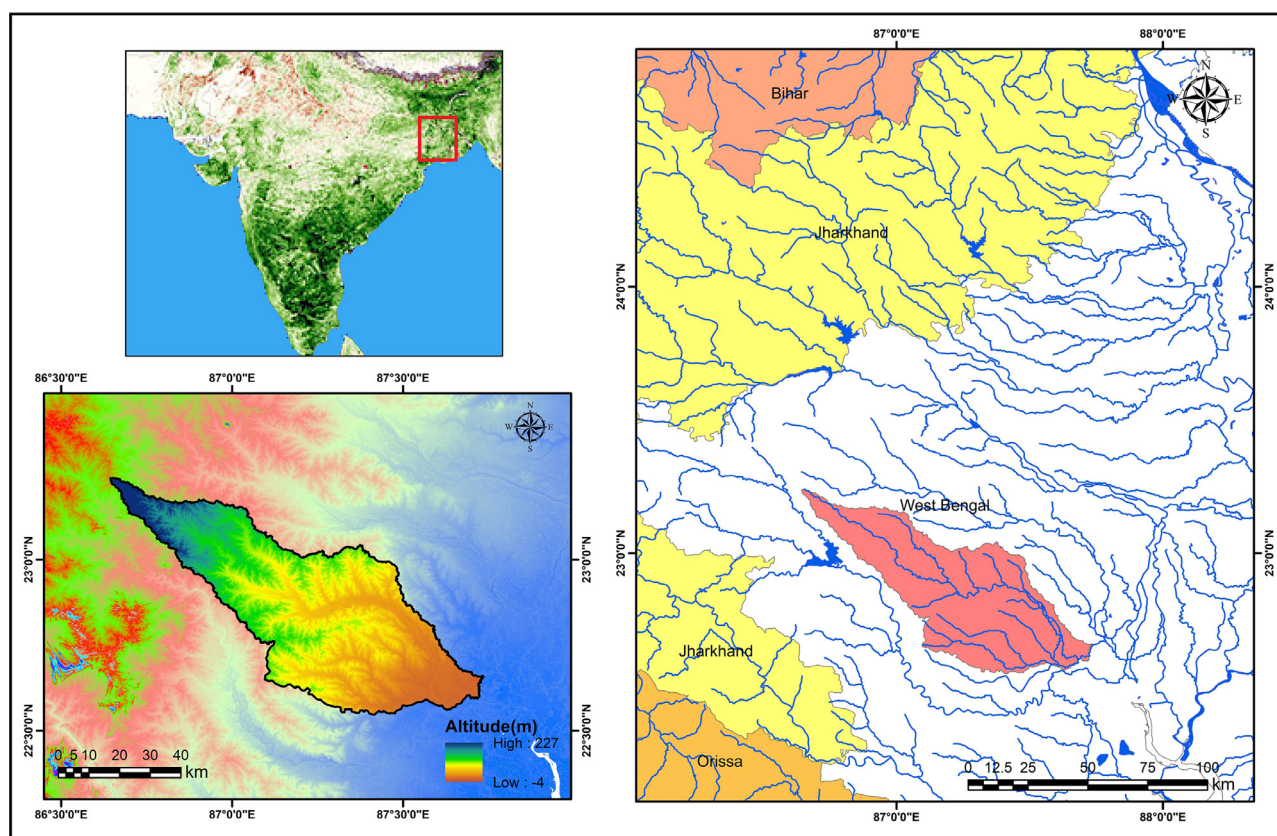


Fig. 1. Geographical location of the study area.

Table 1

Data source and data format of different used thematic layers.

Thematic map layers	Data format	Source of the data
Altitude	Raster grid	Digital Elevation Model from USGS. https://earthexplorer.usgs.gov/
Slope	Raster grid	Digital Elevation Model from USGS. https://earthexplorer.usgs.gov/
Population Density	Raster grid	Census of India data, 2011
Distance from river (Euclidean distance)	Polyline	Digital Elevation Model from USGS. https://earthexplorer.usgs.gov/
Distance from road (Euclidean distance)	Polyline	Google Earth Pro
Agricultural density	Raster grid	Landsat images (TM&OLI) from USGS https://earthexplorer.usgs.gov/
Settlement density	Raster grid	Landsat images (TM&OLI) from USGS https://earthexplorer.usgs.gov/
Forest density	Raster grid	Landsat images (TM&OLI) from USGS https://earthexplorer.usgs.gov/

prepared by Arc GIS 10.3 software and obtained from satellite images (Table 2). The accuracy of the forest maps of both of the years (1998 and 2019) are determined by the error matrix (Table 3). The overall accuracy has been computed by the total corrected and the sum of total pixel in error matrix method. Therefore, it has been stated that both the post classified maps (1998 and 2019) are maintained the accuracy level of 89% and 87% respectively. The forest cover change or the deforested area has been taken place between the years 1998 and 2019. After that a deforestation map with two classes like deforested area (forest to non-forest) and unchanged area (where forest are unchanged) were generated by subtracting the forest cover maps of 2019 from 1998 and it considers as a dependent variable as well as binary variables (Fig. 2).

3.3. Determination of independent variables

Selection of independent variables or the explanatory factors is an important step for deforestation probability modeling. Here, eight fac-

tors (altitude, slope, population density, agricultural density, settlement density, distance from road, and distance from river and forest density) have been selected as dependent variables. Different thematic layers have been prepared for different explanatory factors (Figs. 3 and 4). Deforestation is highly correlated with slope that is a key factor for deforestation and in general condition, vegetation cover and the steepness of slope has a negative or an inverse relationship between each other (Kumar et al., 2015; Sahana et al., 2018). Altitude is an another leading factor that influences the probability of deforestation because high altitude zones trigger the process of weathering, landslide and erosion meanwhile the plain land is very much feasible for the growth of human civilization which exaggerates the human encroachment over forest zone (Bayat 2000). An inverse relationship always shows between forest density and deforestation (Bouldin 2008). Vegetation density and road construction is always correlated because road construction exaggerates the process of deforestation that makes forest zone more vulnerable and fragile (Chomitz and Gray 1999). High population growth

Table 2
Details of the used satellite images.

Space craft id	Sensor id	Date	Path	Row	Sun azimuth	Sun elevation	Band details	Spatial resolution (m)
Landsat 8	OLI_TIRS	Oct-2019	139	044	139.49523543	56.75786880	Band 1 = Coastal Aerosol (0.43–0.45 μm)	30
							Band 2 = Blue (0.45–0.51 μm)	30
							Band 3 = Green (0.53–0.59 μm)	30
							Band 4 = Red (0.64–0.67 μm)	30
							Band 5 = NIR (0.85–0.88 μm)	30
							Band 6 = SWIR 1 (1.57–1.65 μm)	30
							Band 7 = SWIR 2 (2.11–2.29 μm)	30
							Band 8 = Panchromatic (0.50–0.68 μm)	15
							Band 9 = Cirrus (1.36–1.38 μm)	30
							Band 10 = Thermal Infrared 1 (10.6–11.19 μm)	100
							Band 11 = Thermal Infrared 2 (11.5–12.51 μm)	100
Landsat 5	TM	Oct-1998	139	044	136.94502742	51.27926155	Band 1 = Blue (0.45–0.52 μm)	30
							Band 2 = Green (0.52–0.60 μm)	30
							Band 3 = Red (0.63–0.69 μm)	30
							Band 4 = NIR (0.76–0.90 μm)	30
							Band 5 = SWIR 1 (1.55–1.75 μm)	30
							Band 6 = Thermal (10.40–12.50 μm)	120
							Band 7 = SWIR 2 (2.08–2.35 μm)	30

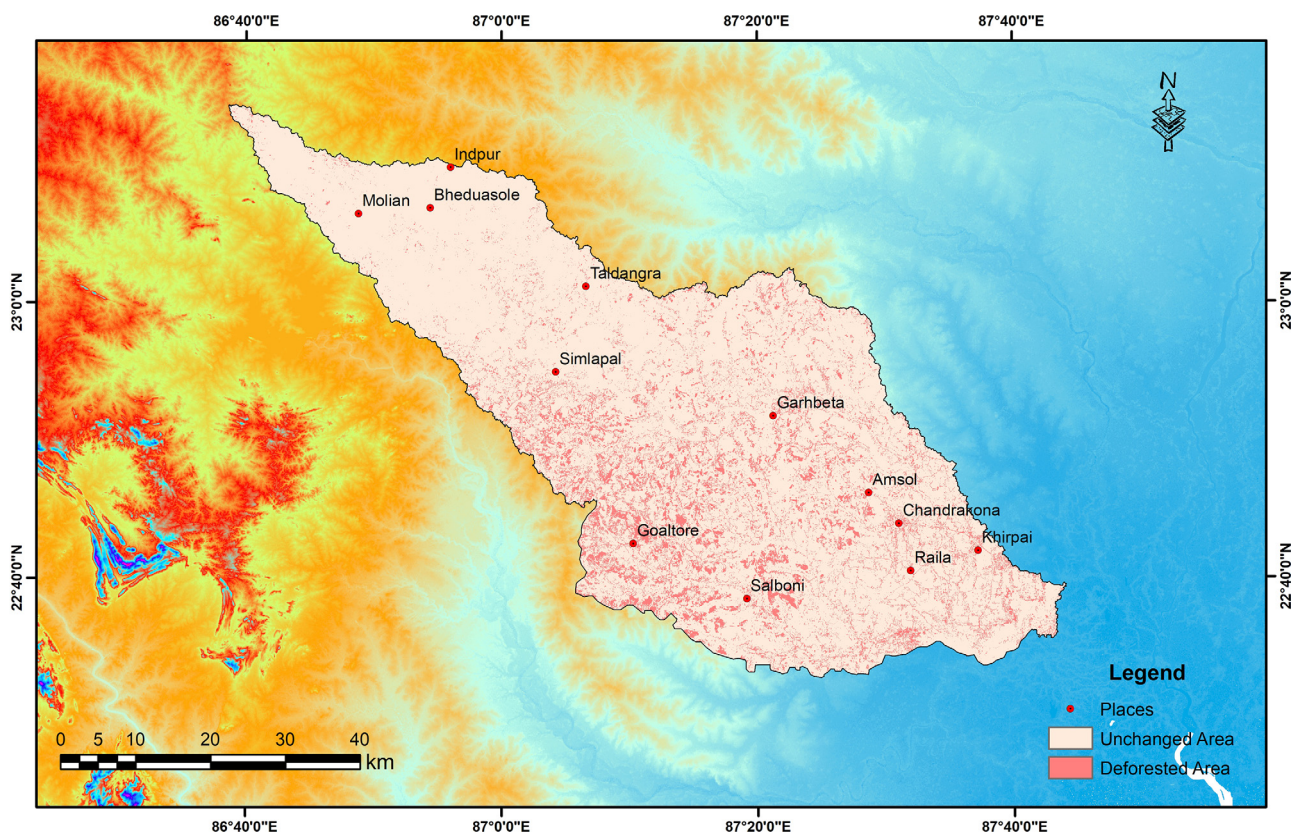


Fig. 2. Thematic layer of deforested map of the study area.

Table 3
Accuracy estimates of forest cover maps (1998 and 2019).

Year	class	Reference data			Accuracy
		Forest	Non-forest	Total	
1998	Forest	61	4	65	Overall accuracy=89%
	Non-forest	7	28	35	
	Total	68	32	100	
2019	Forest	57	6	63	Overall accuracy=87%
	Non-forest	7	30	37	
	Total	64	36	100	

is a crucial factor for deforestation and excessive population pressure with human intervention over a land may cross the boundary of ecological carrying capacity limit (Mather 1991). Agricultural land is another leading explanatory factor of deforestation while the increasing trend of agricultural land is another example of human interference over natural resources (Chakravaty et al., 2012). Distance from river is taken as an explanatory factor here because most of the agricultural lands along with human settlements are flourishing near the river side since the ancient time (Gayen and Saha 2017).

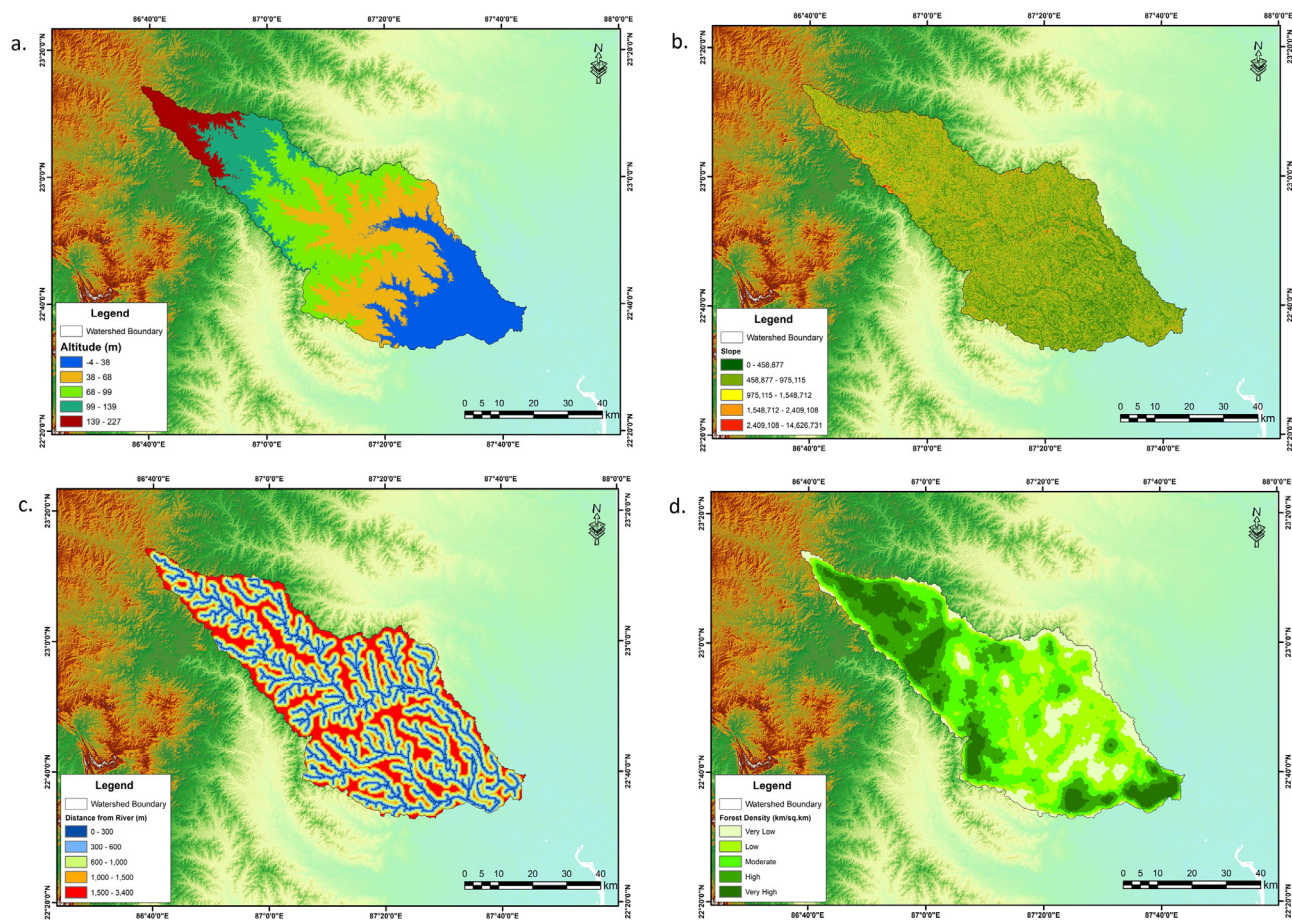


Fig. 3. Different thematic layers of the explanatory factors, a. Altitude b. Slope c. Distance from river d. Forest density.

Table 4
Collinearity statistics of different explanatory factors.

Variables	Collinearity Statistics	
	Tolerance	VIF
Distance from River	.930	1.075
Agricultural Density	.109	4.146
Altitude	.405	2.469
Settlement Density	.247	4.042
Forest Density	.182	3.483
Population Density	.666	1.501
Distance from Road	.878	1.138
Slope	.925	1.081

3.4. Statistical test between variables

In this study, different statistical analysis has been used for better presentation of dependent and independent variables. Eight different explanatory variables have been selected here. The collinearity statistical test has been applied for the validation of the explanatory variables. VIF value >5 and Tolerance value <0.1 also indicate the multi-collinearity problem in the data set (Johnston et al., 2018) (Table 4). Subsequently, another important statistical test like Cramer’s V test has been applied for analysis of the strength of association between dependent variables and the explanatory variables which give a range of 0 to 1 (0 stands for no association whereas 1 stands for complete or full association). This statistical method is based on Pearson’s chi-square statistic. It can test the strength of association between dependent variable and both quantitative and qualitative variables (Kumar et al., 2014). The Cramer’s V

Table 5
showing the association between dependent variable and explanatory variables using Cramer’s V statistics.

Variables	Cramer’s V	P value
Distance from River	0.330	0.00
Agricultural Density	0.832	0.00
Altitude	0.681	0.00
Settlement Density	0.879	0.00
Forest Density	0.822	0.00
Population Density	0.837	0.00
Distance from Road	0.586	0.00
Slope	0.832	0.00

statistic produces V values and P values for all explanatory variables where P stands for probability of the significant level (Table 5). Correlation matrix (using Pearson’s correlation coefficient) has been applied to show relationship among the explanatory factors (Table 6).

Variance Inflation Factor (VIF) and Tolerance have been calculated by using the following equation,

Tolerance = 1 – R²J

VIF = (1 / Tolerance)

R²J indicates the co-efficient of determination of the regression equation of vectors

The chi square statistics for Cramer’s V $x^2 = \sum_{ij} \frac{(n_{ij} - \frac{n_i \cdot n_j}{n})^2}{\frac{n_i \cdot n_j}{n}}$

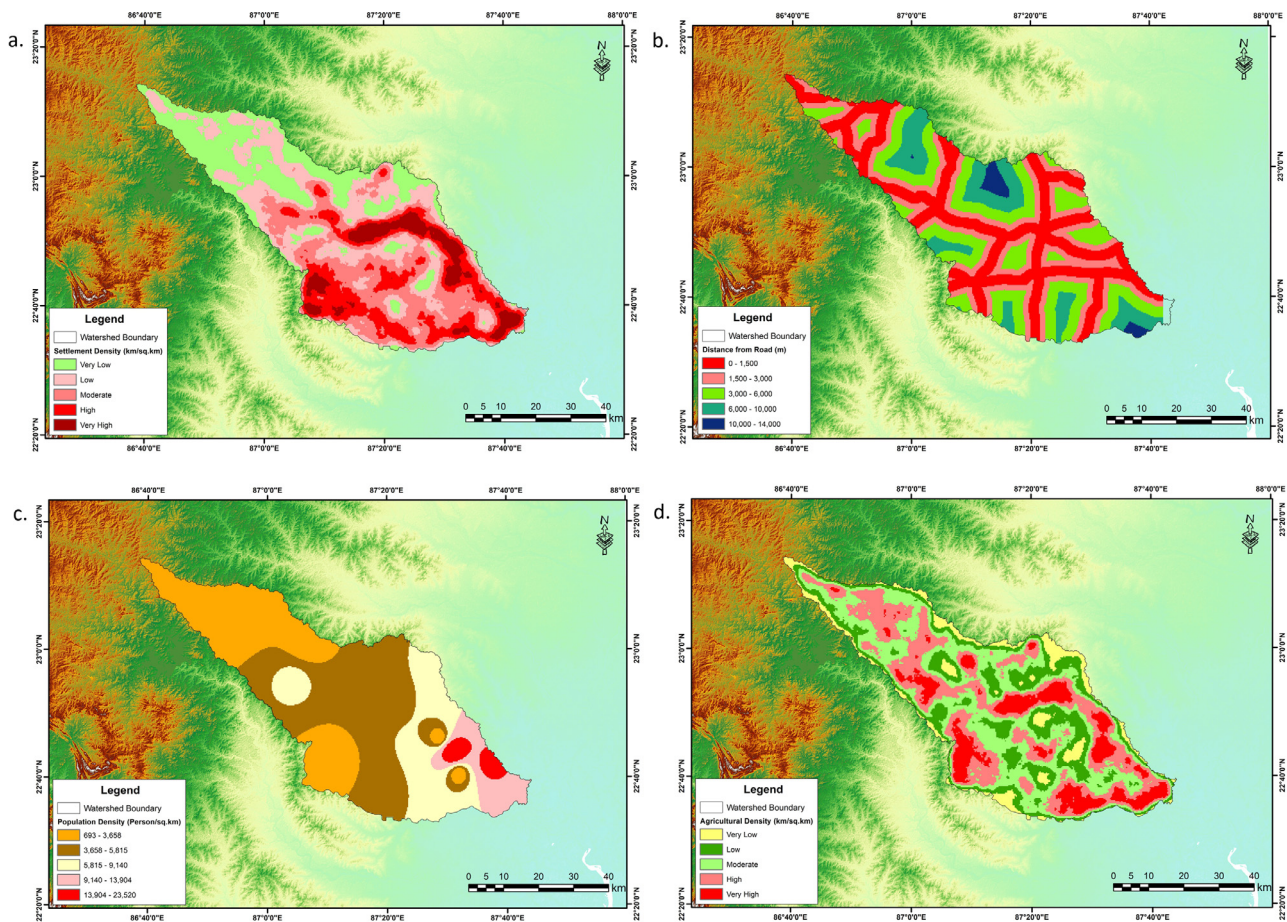


Fig. 4. Different thematic layers of the explanatory factors, a. Settlement density b. Distance from road c. Population density d. Agricultural density.

Table 6

Correlation matrix table shows the relationship among explanatory variables using Pearson's correlation coefficient method.

Variables	Distance from river	Agricultural density	Altitude	Settlement density	Forest density	Population density	Distance from road	Slope
Distance from river	1.000							
Agricultural density	−0.107	1.000						
Altitude	.113	−0.094	1.000					
Settlement density	−0.029	.420	−0.480	1.000				
Forest density	−0.083	.525	.245	.037	1.000			
Population density	−0.051	−0.077	−0.485	.209	−0.264	1.000		
Distance from road	−0.076	.028	−0.195	.082	−0.041	.208	1.000	
Slope	.018	.039	.136	−0.078	.078	−0.109	−0.018	1.000

$$\text{The creamer's V statistics } v = \sqrt{\frac{\varphi^2}{\min(k-1, r-1)}} = \sqrt{\frac{x^2/n}{\min(k-1, r-1)}}$$

Where, φ means Phi-coefficient, X^2 stands for Pearson chi-square test, n signifies total observation, k stands no. of columns and r means no of rows.

3.5. Binary logistic regression model

Binary logistic regression model has been used to analyze the relationship between the dependent and independent variables (Lee 2004). Binary logistic regression model has been successfully applied for the demarcation of deforestation probable zones (Saha et al., 2020). In this research, the model has been fit for better extraction of deforestation probable areas in Silabati watershed. Meanwhile, dependent variables have 0 and 1 binary values (0 value reflects non-deforested zone whereas 1 value reflects the deforested zones). Random point selection technique has been used here for the collection of deforested and non-deforested points and total of around 602 sites have been selected randomly. When

the dependent variable has a binary value then BLR model is suitable for application (Atkinson and Massari 1998). This model also intertwines the maximum likelihood estimation when the dependent variables convert into logit variables (Dai and Lee 2002). BLR model follows-

$$y = a + b_1x_1 + b_2x_2 + b_3x_3 + \dots + b_mx_m$$

$$y = \log_e \left[\frac{P}{1-P} \right] = \text{logit}(P)$$

$$P = \frac{e^y}{1 + e^y}$$

Where, x_1, x_2, x_3 are the causative variable, b_1, b_2, b_3 are the coefficient, a stands for constant value, P represents the probability. Functional y represents the logit (P), y is the associated function of causative factors. All mathematical calculation has been done using statistical package for social science (SPSS v17) software. Deforestation probable map has been prepared by the calculated value of binary logistic regression model using the Arc GIS 10.3 software (Fig. 6).

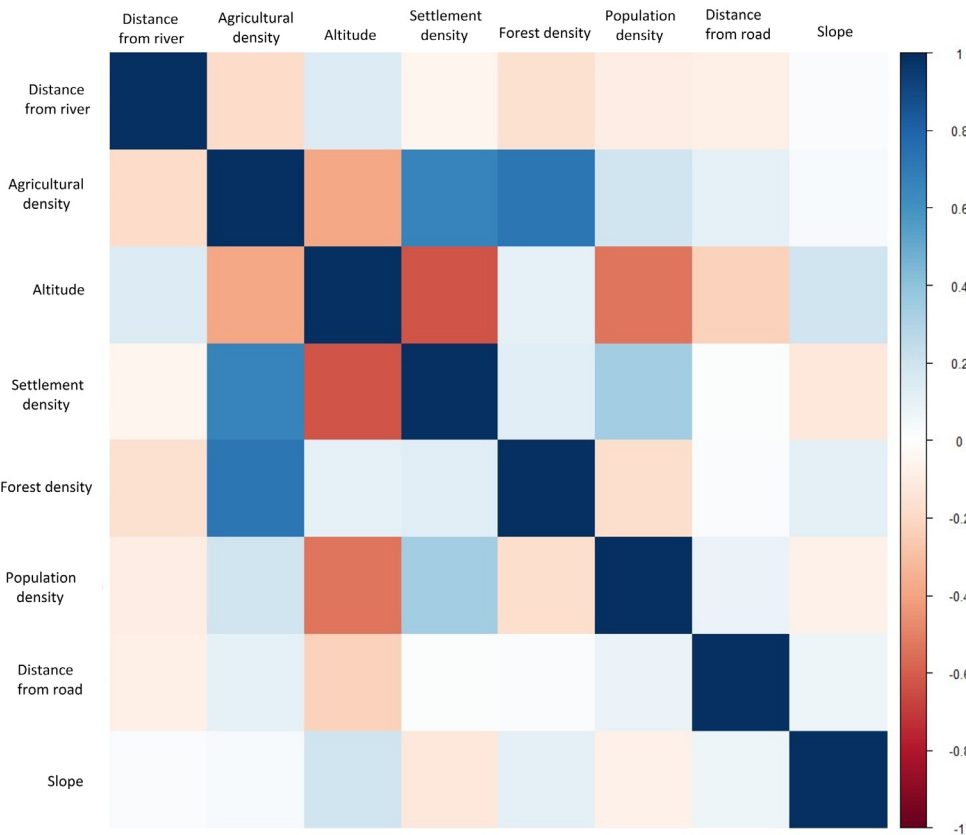


Fig. 5. Correlation among the explanatory factors which reflect by a color matrix scale.

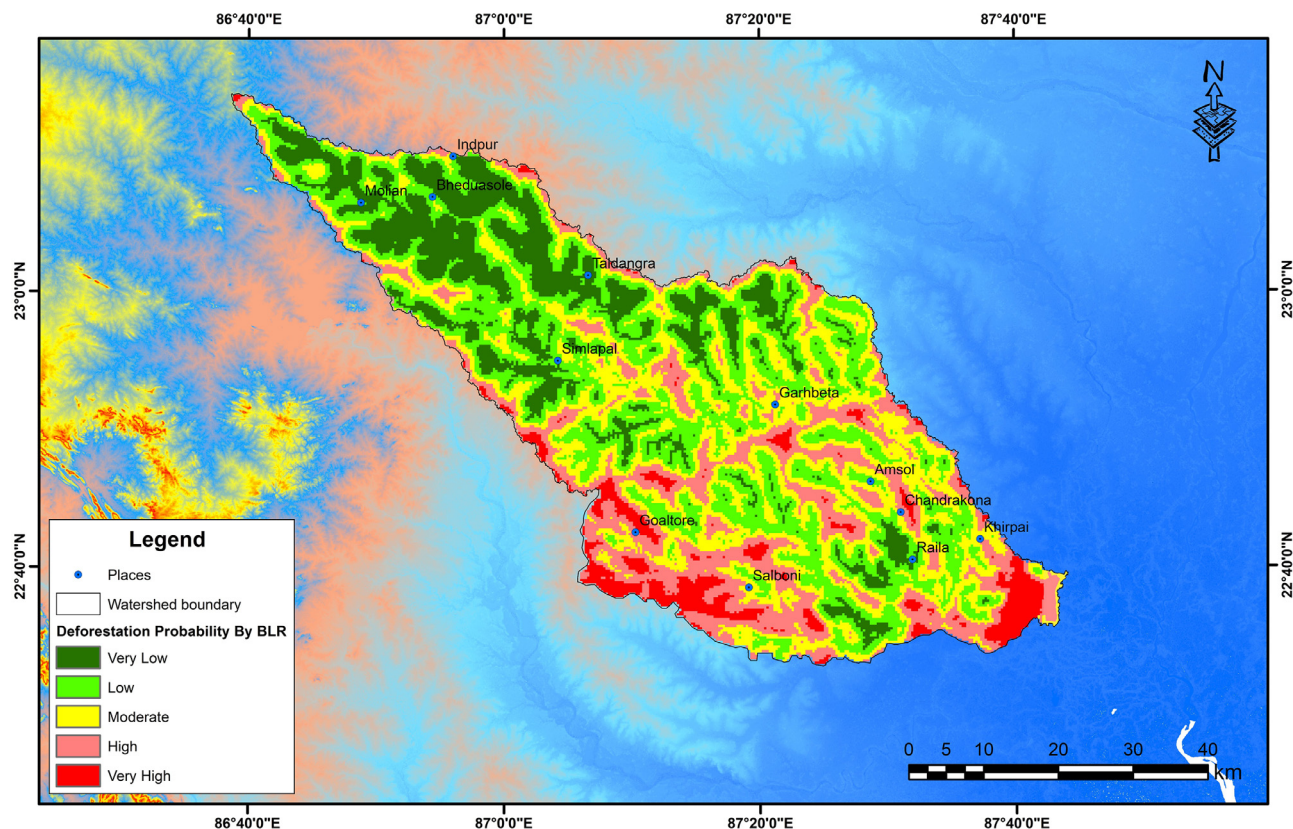


Fig. 6. Deforestation probability map using binary logistic regression model.

Table 7
Hosmer and Lemeshow Test and model summary of accuracy.

Chi-square	df	Sig.	–2 Log likelihood	Cox & Snell R Square	Nagelkerke R Square
18.843	8	.16	649.543	.265	.353

Table 8
Result of binary logistic regression model with overall statistical explanation.

Explanatory variables	B	S.E.	Wald	df	Sig.	Exp(B)	95% C.I. for EXP(B)	
Agricultural density	.001	.000	14.093	1	.000	1.001	1.000	1.001
Distance from river	–0.208	.044	22.733	1	.000	.813	.746	.885
Altitude	–0.004	.003	1.930	1	.16	.996	.989	1.002
Settlement density	.175	.028	39.179	1	.000	1.192	1.128	1.259
Population density	.153	.043	12.451	1	.000	1.165	1.070	1.268
Forest density	.006	.000	4.478	1	.034	1.000	1.000	1.000
Distance from Road	.005	.000	47.663	1	.000	1.000	.999	1.000
Slope	.009	.000	18.598	1	.000	1.000	1.000	1.000
Constant	1.589	.813	3.820	1	.051	4.898		

Table 9
Classification summary of the logistic regression model.

Observed deforested area	0	1	Corrected percentage
Predicted deforested area			
0	231	67	0.7751
1	86	217	0.7161
Overall corrected percentage	–	–	0.7456

4. Results

4.1. Statistical analysis for the significance of the model

Collinearity analysis also reveals that there is no multi collinearity problem in the variables as all the variables have a less than 5 VIF value and greater than 0.1 Tolerance value (Table 4) which portrays that all the variables are independent and ready for the prediction (Saha et al., 2020). Cramer's V test has been done to represent the association between explanatory variables. The strength of association between deforestation and explanatory variables is also shown in Table 5. The density of agriculture, settlement, forest, population and slope have a high association with deforestation whereas distance from river, distance from road, altitude have low association in compare with above mentioned variables. The correlation matrix (using Pearson's correlation method) shows the relationship among the explanatory variables (Table 6).

4.2. Deforestation probability zonation applying BLR model

The final map has been categorized into five different classes such as very low (597.56 sq.km or 17.63%), low (1034.52 sq.km or 30.52%), moderate (939.61 sq.km or 27.72%), high (607.77 sq.km or 17.93%), and very high (210.54 sq.km or 6.21%) applying BLR model. Relevant statistical method such as Hosmer-Lemeshow test, Cox and Snell R^2 and –2log likelihood have been applied to show the integrity of fitness of the model (Table 7). If the significance value of chi-square is more than 0.05 in Hosmer-Lemeshow test than the model consider as good. In case of this study, the significance of chi-square value is 0.16 which shows that the model is significant. Other statistical tests such as Cox and Snell R^2 and Nagelkerke R^2 have also been used here. The R^2 values of both tests are 0.265 and 0.353 respectively that indicate the model is relatively good. The calculated co-efficient values also divulge that all independent variables have different controlling factor over deforestation.

5. Discussions

Deforestation probability mapping is a scientific tool that identifies the future deforested areas and considering the model forest conservators can take sustainable management plan (Lambin 1994). Remote sensing (RS) and Geographical Information System (GIS) techniques along with proper statistical probabilistic method make the model more realistic and scientific. The model based future deforestation map exhibits that the largest part of the deforestation will occur at the middle and middle lower part of the Silabati watershed (Fig. 6). Within the watershed, important forest ranges include Arabari forest range, Salbari forest range, Goaltore reserve forest, Hoomgarh forest range, Gorhbeta forest range etc. In the last three or four decades, the various human actions such as natural forest habitat destruction for shelter, conversion of forest land into agriculture, execution of different developmental projects within the forest and forest peripheral zones etc. have been tremendously amplified in different tropical and sub-tropical forests pockets of Indian sub-continent (Chamling and Bera, 2020a). The large scale bio diversity depletion, man animal conflict, poaching, timbers trafficking etc. are rampant incidents in most of the tropical countries of the world due to lack of proper monitoring and paucity of implementation of laws (Chamling and Bera, 2020b). In general, the degree of regional development has been taken place around the growth poles. In case of Silabati watershed, most of the regional growth centers are confined within forest covered area and sometime two or three sites are restricted by forest cover. In that situation, government has executed differential developmental projects within the forest zones considering the demand of the local people. In the last two decades, magnitude of anthropogenic stresses has been intensified in the different pockets within the watershed. The high decadal population growth of PaschimMednapore (13.86%), Bankura (12.64%) and Purulia (15.43%) has been recorded in 2011 Census of India. People who are residing within this watershed and adjoining districts are encroaching illegally and forestland is being converted into agricultural land in rapid rate. In the recent years, various governmental and nongovernmental projects have also been executed in different parts of the watershed by clearing the forests. The present model of deforestation is highly influenced by different aspects like settlement density, population density and distance from river. When settlement and population density are increased it also leads the deforestation process that means it has a positively correlation whereas negative correlation shows between deforestation and distance from river. In this model agricultural density, forest density, distance from road and slope have a positive correlation with deforestation meanwhile altitude has a negative correlation with deforestation. Very low and low deforestation probability areas have been observed

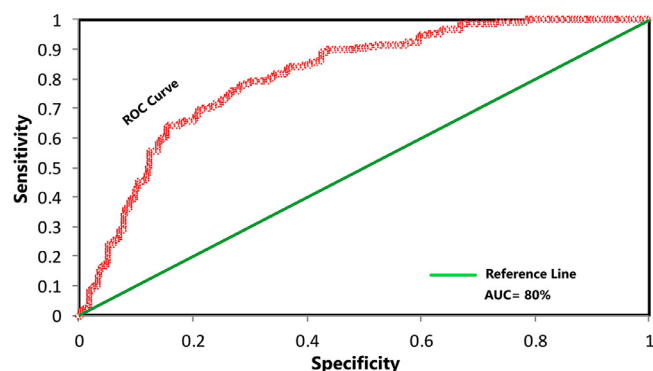


Fig. 7. ROC curve for validation of the model.

mainly in the upper part of the basin (Fig. 6) which contains around 48.15% of the total basin area. High and very high probability zones has been observed in middle and lower middle part of the basin (Fig. 6) which holds total of 24.14% area. This area is highly deforestation probable zone due to existence of high population density, high agricultural intensity and expansion of transport network system. The pattern of ROC curve and the value of AUC give a satisfactory result with high accuracy from this model. The ROC curve represents an upper left pattern that makes the model more reliable while the value of Area Under Curve is 0.80, that makes the model more valid and applicable.

6. Model validation

Without proper validation, any scientific model is practically irrelevant; every scientific model needs acceptance and accuracy. Various predicted research models such as landslide susceptibility zonation, deforestation probable zone identification, spring potentiality and ground water potentiality mapping are properly validated by the Receiver Operating Characteristics (ROC) curve. The ROC curve has been developed graphically when True Positive Rate (TPR) or sensitivity values is in Y axis whereas False Positive Rate (FPR) or specificity is in X axis with a selected cut-off threshold. High sensitivity recommends high accuracy whereas high specificity suggests lower accuracy of the model. The value of Area Under Curve (AUC) is another important measurement of the model validation.

TPR and FPR are as follows:

$$TPR = \frac{TP}{TP + FN}$$

$$FPR = \frac{FP}{FP + TN}$$

Where, TN stands True negative, TP means True positive, FN presents False negative and FP stands False positive.

Area Under Curve (AUC) is another quantitative explanation for model validation. The value of AUC lies in between 0 and 1. Similarly, the values of (0.9–1), (0.8–0.9), (0.7–0.8), (0.6–0.7), (0.5–0.6) AUC represent perfect, very good, good, moderate and weak respectively for model validation. Quantitatively, validation of any scientific model is analyzed by ROC/AUC curve (Fig. 7). Basically, ROC curve measures the accuracy level of the model. The ROC curve is generated here by the predicted and actual forest cover areas of Silabati watershed. The value of AUC (Area Under Curve) also reveals the accuracy of the model. In this model Area Under ROC Curve is 0.801 that means this model also gives 80% accuracy and all the independent factors are taken a significant role in forest conversion. The Boolean ROC plot on the basis of sensitivity against the specificity also provides us a visual validation of the model. If the plot is closer to the northwest corner or upper left corner of the chart area then it indicates the overall accuracy of the model. In this study ROC plotting also reveals that the model is statistically significant due to its North West direction in the plot area.

7. Conclusions

We live in the era of steady forest land degradation, every day forest lands are being destroyed due to anthropogenic process which is a concerned topic to researchers in the present day context. Deforestation probability mapping is a tool which may help us to restore the forest areas again and that encourage the researchers to apply the model in forest degraded regions of various parts in the world. Deforestation is a burning issue in India along with the whole tropical countries in the world. Forest land within Silabati watershed is being degraded very fast in the last few decades. The deforestation probability map also shows that the middle and lower middle part of the basin is more vulnerable. According to forest survey report 1985, the dominated plant species in the lower middle and middle part is Sal (*Shorea robusta*) along with other species such as *Pterocarpus marsupium*, *Terminalia tomentosa*, *Bombaxceiba*, *Adina cardifolia*, *Buchanania latifolia* etc. which have a great economic importance that makes the forest regions more fragile. Explicitly, diverse human activities like unplanned settlement growths, infrastructural expansion (like road, railway, school, college, government offices etc.) are principal regulating factors which accelerate the likelihood of deforestation within Silabati watershed. But now state government also takes some steps to recover the forest region and protects (for example Arabari Joint Forest Management) the deforestation probable zones. It is also the government's responsibility to introduce different sustainable management plan for protection of forest habitats along with wild species. The deforestation probability map may help the forest officials to take action over deforestation hot spot zones in different vulnerable pockets within the study area. Before the execution of any sustainable forest management plan along with development for livelihood of forest dwellers, this rational and scientific model can implement to assess the likelihood of deforestation of any region of the world. Subsequently, more scientific research and development are required to make suitable management strategies for the conservation of biotic resources as well as forest habitats.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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